

Harshit Garg

Gaithersburg, MD

garg.archie@gmail.com | (301) 525-7100

www.harshit-garg.com
<https://github.com/hgarg1>

Selected Articles/Publications:

- <https://cmns.umd.edu/news-events/news/cracking-cryptography-classroom>
- <https://www-math.umd.edu/about-us/math-news.html?id=139&start=30>

Technical Proficiencies:

Programming Languages

- **Core & Backend:** Java, Python, C, C#, C++
- **Functional & Scripting:** Ruby, R, OCaml

Backend, Web & Mobile Frameworks:

- **Backend:** Spring Boot, ASP.NET Web API
- **Full-Stack:** Next.js, Nest.js, Vaadin, JSP
- **Mobile Dev:** Jetpack Compose, Flutter

Databases, ORMs & Caching:

- **Databases:** MySQL, SQL Server, PostgreSQL
- **ORMs:** Prisma, Entity Framework (EF)
- **Caching & Streaming:** Redis, Apache Kafka

Data Science, Machine Learning & Analytics:

- **Tooling:** PyTorch, ML.NET, Jupyter, MATLAB

Integrations & Ecosystems:

- **Payment Gateways:** Stripe / FinTech APIs
- **Cloud, DevOps & Infrastructure:**
- **Platforms & Tools:** Docker, Azure DevOps, Git
- **Networking & Protocols:** TCP/TLS

Professional Certifications:

- AWS Certified Cloud Practitioner (CLF-C01)
- Microsoft Azure Fundamentals (AZ-900)
- CompTIA A+ Certified

Core Competencies:

- Agentic AI/Workflows, Multi Agent Systems, NLP/BERT, Eval Pipelines, Neural Networks
- Event-Driven Distributed Systems, Deep Cloud Architecture, Platform Engineering, Cryptography
- Advanced Data Structures, Design Patterns, Agile & Scrum, 6 Sigma

Leadership

- Member, **Microsoft** Customer Advisory Board
- Founder and Former President – Collegiate FBLA at UMD and Collegiate DECA at UMD
- Former President, College Park Toastmasters
- Former **Senator**, University of Maryland Senate.
- **Current** Chief Judge with the Montgomery County Board of Elections.
- Former Delegation Chair, Board Chair, and **Speaker Pro Tempore** – Maryland Student Legislature at UMD

EXPERIENCE

Software AI Intern – Higher Logic

February 2025 to May 2025

- Architected and deployed a self-healing production system, integrating **DataDog** observability, **Devin** autonomous agents, and Slack/Microsoft Teams to automatically detect, triage, and remediate recurring application failures within **30 min**.
- Designed log consolidation and root-cause intelligence pipelines that materially reduced time-to-resolution (TTR) by **25%** and improved engineering throughput overall.
- Collaborated directly with the CTO and Senior Director to integrate the system into acquired legacy **enterprise platforms**, ensuring compatibility with existing infrastructure.
- Developed strategic understanding of acquisition-driven platform consolidation and **enterprise software integration** at scale across **300 employees**.

IT Intern & Co-op – United Parcel Service (UPS: ENT & OPT Divisions)

June 2023 to October 2024

- Architected and maintained enterprise BI dashboards (Google Looker) over multi-terabyte datasets, supporting a **\$750K** GCP budget and implementing IAM-based automated provisioning using **BigQuery** and on premises **linux environments**.
- Integrated time-series **forecasting**, data science models into **production analytics** workflows; contributed to HR Power BI dashboards improving decision making by **50%**
- Founded hackathon team that **placed 3rd** nationally for a carbon neutrality solution; **received recognition** from Senior Director, CTO, and a CO-OP offer.
- Help develop AI assistant **enabling complex** package management for **official UPS app** across **1.1M Users** using Spring Boot backend plus **LLM APIs** to **personalize UX**

Web Assistant & Technical Advisor – UMD CMNS Dean's Office

September 2022 to May 2024

- Sole student developer responsible for maintaining and enhancing UMD CMNS websites using Drupal, RESTful APIs, and Google Maps API-based **data visualizations**.
- Integrated external data sources into consistent, production-ready web interfaces via REST APIs to accomplish basic **Retrieval Augmented Generation**
- Used Python (Google Colab) and Excel libraries to streamline mailing list infrastructure, reducing operational costs by **\$1,000s** of dollars and optimizing spreadsheet storage efficiency.
- Advised Assistant Dean of Communications and Senior Leadership to empower **6+ initiatives** across **10 departments** improving outcomes for **20+ thousand students**

SOFTWARE ENGINEER – Maryland Technology Enterprise Institute

November 2021 to August 2022

- Sole student engineer within a two-person IT department, developing full-stack ASP.NET Web API systems **supporting** the School of Engineering across **6,000 students**.
- Re-architected legacy applications into modular, POCO-driven architectures using advanced LINQ and **optimized SQL**, applying clean architecture and separation-of-concerns principles to improve scalability and maintainability.
- Managed CI/CD pipelines, builds, and deployments via Azure DevOps, ensuring reliable **release workflows**.

SOFTWARE DEVELOPER Intern – UNIONTRACK

October 2019 to August 2021

- Expanded expertise across distributed systems and emerging technologies, including SQL/NoSQL design tradeoffs, blockchain fundamentals, and Python ML/AI tooling.

EDUCATION

Bachelor of Science, Computer Science (MACHINE LEARNING TRACK). Minors: Technology Entrepreneurship & Business Analytics

University of Maryland, College Park, MD: August 2021 - December 2025

- **Winner**, Synergy in **Data & AI** Award — UMD Department of Computer Science
- Completed **400-level coursework** beginning freshman year
- Dean's List (2021–2022)
- Hinman CEOs Program · Terrapin Leadership Institute (TLI)
- Chief Executive Officer, Data and Cloud Computing Society (**300+ Members**)
- Accepted into IT and Business Transformation Fellowship
- Selected coursework: Ethics in Technology, Advanced Data Analysis, Computer Vision

ACHIEVEMENTS AND ACCOMPLISHMENTS AND PROJECTS

Competed in the 2026 Ernst and Young AI Challenge (water scarcity)

- PyTorch, complex feature engineering, ensemble modeling.
- Conducting primary stakeholder interviews with communities in water-scarce African regions to ground the solution in real domain knowledge and improve model relevance

1. Lead Agentic AI Engineer – PTA.AI (<https://pta.ai.harshit-garg.com/>):

- Architected a **multi-tenant SaaS platform** for PTA and school websites with environment-aware configuration (dev/prod), tenant isolation, and role-based access control which meant **RBAC-driven admin and tenant dashboards**, enabling scoped permissions, delegated user management, and secure configuration changes
- Architected and developed a **distributed** tenant governance architecture **with multi-parent approval workflows**, staged policy execution, and hierarchical control across organizations, enabling secure **multi-organization** management.
- Built a **dynamic page and content generation system** backed by a database, allowing non-technical users to manage events, pages, and site content without redeploy. This follows the **builder, auditor 1, ci/cd builder 1, auditor 2** pipeline and state graph
- Integrated **third-party services** (domains, payments, email, AI inference) with backend orchestration and frontend safety checks such as Namecheap domain management API and GitHub organization and repository management via installed GitHub application.
- Owned end-to-end system design including **backend APIs, database schema, auth flows, deployment pipelines, and cloud infrastructure**
- Built and operated a full-stack ecosystem including **web, desktop, and mobile interfaces** with automated CI/CD deployments (**EXPO**) and cloud-hosted infrastructure to automate desktop and mobile applications releases.

2. Lead ML Engineer - LLM Arena (<https://llm-arena.harshit-garg.com/>):

- Designed and built a **competitive evaluation platform for large language models**, centered on adversarial, head-to-head benchmarking
- Implemented **structured evaluation pipelines** to run controlled prompt scenarios, capture model outputs, and score performance across multiple dimensions
- Engineered reproducible benchmarking workflows emphasizing **determinism, isolation, and auditability** rather than leaderboard overfitting
- Built abstractions for **model adapters**, allowing consistent evaluation across heterogeneous LLM APIs
- Led system architecture with a focus on **fairness, replicability, and longitudinal comparison** of model behavior

3. Lead AI Engineer – Campaign Sites (<https://campaign-sites.harshit-garg.com/>):

- Architected and launched a **multi-tenant campaign website platform** enabling political candidates and organizations to rapidly deploy secure, customizable campaign sites. Engineered a **mobile-first full-stack architecture** supporting web, desktop (Electron), and mobile clients with shared authentication and backend services.
- Implemented **secure authentication, role-based access control, and supporter management systems** to handle campaign staff, volunteers, and public users.
- Designed and maintained **automated CI/CD pipelines with GitHub Actions** to build, test, and deploy applications across environments with minimal operational overhead.
- Developed **scalable backend services and data models** supporting campaign messaging, supporter engagement workflows, and analytics.

4. Lead Software Engineer – Multitenant Email Based Support and RBAC Platform (<https://rbac-system.archiegarg.me>)

- Designed and built a **centralized, multitenant email delivery platform** enabling multiple independent applications to send transactional and system emails through **service accounts** with tenant-specific branding, language, and sender identity
- Implemented a **shared RBAC control layer** consumed by downstream services, allowing applications to either **inherit platform-defined permissions** or map their own role models into a unified authorization schema
- Built **RBAC-as-a-Service (RBACaaS)** abstractions to standardize permissions, roles, and access scopes across heterogeneous applications without duplicating authorization logic
- Enforced **tenant isolation and policy enforcement** while supporting cross-application governance, auditability, and consistent access semantics

5. Lead AI Systems Engineer – Nexus Intelligence (<https://nexus-intelligence.harshit-garg.com/>):

- Architected and implemented a **full-stack AI orchestration platform** extending Google Gemini into a persistent, version-controlled intelligence system with mutable memory, branching, and checkpoints.

- Designed a **Git-inspired version control model for AI conversations**, including branches, checkpoints, pull requests, and compile steps to materialize model state deterministically.
 - Built a **Memory Vault subsystem** enabling durable, queryable AI memories with scoped visibility, context injection, filtering, and explicit add/remove protocols.
 - Implemented a **real-time command center** exposing system telemetry (latency, uptime, auth health), active agents, channels, and neural engine state.
 - Developed a **multi-agent registry and orchestration hub**, discovery, initialization, and deployment of specialized AI agents across productivity, coding, creative, and utility domains.
 - Engineered **secure RBAC and organization-aware access control**: allowing, supporting users, admins, teams, org memberships, and sector-based permissions.
 - Created a **low-latency AI execution pipeline** with configurable inference parameters (temperature, top-p, max tokens) and runtime engine health monitoring.
 - Built a **collaboration layer** (Collab Nexus) with real-time channels, squads, announcements, and DevOps communication integrated directly into the AI platform.
 - Implemented **secure authentication flows** including passkeys, biometrics, hardware keys, and system override credentials.
 - Designed and shipped a **high-fidelity UI system** with modular panels, live state indicators, and responsive layouts optimized for complex workflows.
 - Owned the platform end-to-end: **frontend architecture, backend services, database design, auth, infra, and deployment**.
6. Lead Software Engineer – Codex Hotels (<https://hotel.archiegarg.me/>)
- Designed and built a **full-stack hotel booking system** with **stateful booking orchestration**, enforcing step order and validating inputs (dates, capacity, availability) before allowing progression
 - Modeled **room inventory and pricing logic**, including capacity constraints, stay attributes, and per-night rates surfaced dynamically in the UI
 - Engineered a **transaction-style reservation flow** that mirrors real hospitality systems while keeping a **high-polish, futuristic UI** without sacrificing functional correctness, accessibility, or flow integrity
 - Owned end-to-end system design, including frontend architecture, booking state management, and deployment
7. Lead Software Engineer – AI Terminal (<https://ai-terminal.archiegarg.me/>)
- Designed and built an online **AI-augmented PowerShell environment** that integrates directly with VS Code to assist developers during command composition and execution
 - Implemented **pre-execution analysis and feedback**, allowing errors, misconfigurations, and risky commands to be identified before runtime
 - Built deep editor–terminal integration to provide **context-aware suggestions**, explanations, and corrective guidance inline with developer workflows
 - Enabled **version-control-aware command execution**, allowing the system to reason about build, run, and deployment commands in source-controlled environment.
 - Focused on **shift-left developer experience**, catching impediments earlier in the development lifecycle to reduce iteration cost and runtime failures
 - Owned end-to-end system design, including terminal hooks, editor integration, AI interaction layer, and execution orchestration
8. Desktop Apps: Grok Client, OpenAI Java Client, Dream X Client, Google Gemini Nexus Client (Personal Projects)
- Rebuilt LLM interface in Java with Gmail API auth, image uploads, markdown rendering
 - Signed EXE binaries and packaged via Inno Setup with clean install/uninstall lifecycle
 - Implemented secure credential storage using Windows Credential Manager
 - Designed OTA update mechanism via signed GitHub releases
 - Used latest Java (MigLayout and FlatLaf), Node.js (React.js, Next.js, Nest.js), and Python (pytorch, cuda) libraries

Throughout all projects:

- Learned the limitations and strengths of Devin / Cursor / Claude Code / Codex Cli / GEMINI CLI via **hundreds of hours** of use